

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	1 June 2026
Team ID	SWTID-2026-8236
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	• Register using application form • Login using Email & Password • Secure user authentication
FR-2	User Authentication	• User login validation • Forgot Password (Optional) • Session management
FR-3	Credit Card Application	• Fill applicant details (Income, Age, Occupation, Credit History, etc.) • Validate mandatory fields • Save application details in database
FR-4	Data Preprocessing	• Handle missing values • Encode categorical data • Normalize input features
FR-5	Credit Card Approval Prediction	• Load trained ML model • Predict Approval/Rejected status • Display prediction probability (confidence score)
FR-6	Prediction Result	• Display approval/rejection result • Show confidence score • Display reason/important factors (optional)

Technologies Used:

Component	Technology
Frontend	HTML5, CSS3, Bootstrap, JavaScript
Backend	Python, Flask
Machine Learning	Scikit-learn, Pandas, NumPy
Database	MySQL
Model	Random Forest / XGBoost
Deployment	Localhost / Render / AWS
IDE	Visual Studio Code
Verson Control	Git & GitHub

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should have a simple, user-friendly interface that allows applicants to submit applications easily.
NFR-2	Security	Protect user data using password hashing, secure authentication, input validation, and HTTPS communication.
NFR-3	Reliability	The prediction system should provide consistent and accurate results with proper error handling.
NFR-4	Performance	Generate prediction results within 2 seconds under normal system load.
NFR-5	Availability	The system should be available 24×7 with regular database backup and recovery support.
NFR-6	Scalability	The application should support an increasing number of users and prediction requests without performance degradation.

